

Differential Summing Stage (OPA1612): Working Document

Digital Audio Player • Rev 5

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1. Summary

Field	Value
Manufacturer / MPN	Texas Instruments, OPA1612AIDRGR (AL-PN #1-104)
Package	8-SON, 3 × 3 mm, dual op-amp; one package covers both channels
Function	Differential summing with a second-order multiple-feedback (MFB) Butterworth low-pass
Signal chain	I/V (×2 per channel) → differential summing and MFB low-pass → OPA1622 output stage → 3.5 mm jack

Each channel's two I/V outputs are subtracted into one single-ended signal, scaled to the headphone level, and low-pass filtered, all in one op-amp. MFB was chosen over Sallen-Key because it can attenuate (gain below 1), subtracts in the same stage, keeps rolling off into the MHz region where Sallen-Key feedthrough limits rejection to roughly 40 to 60 dB, and holds its inputs at a virtual ground, so there is no common-mode input swing on ± 3.3 V rails. A Butterworth response gives the flattest passband with no ripple.

2. Design Requirements

Requirement	Value
Gain	0.44: 0.85 V peak (0.6 V RMS) headphone target ÷ 1.93 V peak nominal differential input. The OPA1622 stage is unity gain.
Response	Butterworth, $Q = 0.7071$, passband ripple 0 dB
Corner frequency	$f_0 = 70$ kHz (minimum 60.96 kHz for ≤ 0.05 dB loss at 20 kHz; margin for tolerance)
Passband loss	Chain total ≤ 0.1 dB at 20 kHz (I/V + summing)
Supply	± 3.3 V analog rails

Requirement	Value
Constraints	Low noise (resistors in the hundreds of Ω) · Small footprint · C0G capacitors in the signal path

3. Pin Configuration

Channel A is left and channel B is right. Component labels follow the TI cookbook: R3 input, R1 feedback, R2 middle, C1 node-to-ground, C2 feedback.

Pin	Name	Connection
1	OUT A	SUMM_L; R1 (220 Ω) back to node X_L ; C2 (2.7 nF) back to -IN A
2	-IN A	From IV_L via R3 (499 Ω) $\rightarrow$ node X_L (C1 13 nF to GND) $\rightarrow$ R2 (681 Ω)
3	+IN A	From IV_LB via the mirrored network: R3, node (C1 and R1 to GND), R2; C2 from +IN A to GND
4	V-	-3V3, 0.1 μ F X7R to GND
5	+IN B	From IV_RB via the mirrored network
6	-IN B	From IV_R via R3 $\rightarrow$ node X_R (C1 to GND) $\rightarrow$ R2
7	OUT B	SUMM_R; R1 back to node X_R ; C2 back to -IN B
8	V+	+3V3_ANALOG, 0.1 μ F X7R to GND

Polarity: the I/V stage inverts ($IV_L = -k \cdot s$, $IV_{LB} = +k \cdot s$), so the output is $G \times (k \cdot s - (-k \cdot s)) = +2Gk \cdot s$, in phase with the audio. This assumes DACL and DACR are the true-phase DAC outputs.

4. External Components

Ref	Value	Part / Notes	Qty	AL-PN
U	Dual op-amp	TI OPA1612AIDRGR, 8-SON	1	#1-104
R3	499 Ω , 0.1%, 0402	YAGEO RT0402BRD07499RL, thin film. Same part as I/V R_f .	4	#1-105
R1	220 Ω , 0.1%, 0402	YAGEO RT0402BRD07220RL, thin film. Feedback and mirrored ground leg.	4	#1-109
R2	681 Ω , 0.1%, 0402	YAGEO RT0402BRD07681RL, thin film	4	#1-110
C1	13 nF, 50 V, C0G, 5%, 0805	Murata GRM2195C1H133JA01D. One per X node; also filters common-mode.	4	#1-108
C2	2.7 nF, 50 V, C0G, 2%, 0402	Murata GRM1555C1H272GE01D. Feedback and mirrored ground leg.	4	#1-107
C_{dec}	0.1 μ F X7R, 0402	One per supply pin	2	

5. Equations and Results

Transfer function (TI labels)

$$H(s) = \frac{1}{s^2 + s \frac{1}{C_1} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

Source: [Analog Engineer's Circuit Cookbook: Amplifiers, MFB Low-Pass Filter \(Texas Instruments\)](#).

Design equations (matching the denominator to $s^2 + (\omega_0/Q)s + \omega_0^2$)

$$G = \frac{R_1}{R_3} = \frac{220}{499} = 0.441$$

$$C_1 = \frac{Q}{\omega_0} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) = 12.89 \text{ nF}$$

$$C_2 = \frac{1}{\omega_0^2 R_1 R_2 C_1} = 2.68 \text{ nF} \quad (\omega_0 = 2\pi \cdot 70 \text{ kHz}, Q = 0.7071, R_2 = 681 \Omega)$$

Source: [Analog Engineer's Circuit Cookbook: Amplifiers, MFB Low-Pass Filter \(Texas Instruments\)](#); derived from the transfer function above.

Passband loss (general second-order form, $x = f/f_0$)

$$|H|^{-2} = (1 - x^2)^2 + \left(\frac{x}{Q}\right)^2, \quad \text{Loss} = 10 \log_{10} |H|^{-2}$$

Source: standard second-order low-pass magnitude; reduces to $1 + x^4$ at $Q = 0.7071$.

Quantity (standard values 13 nF, 2.7 nF)	Value
f_0 / Q	69.4 kHz / 0.707
Summing-stage loss at 20 kHz	0.030 dB
Chain loss at 20 kHz (with I/V 0.038 dB)	0.068 dB
Attenuation at 1 MHz: summing stage / whole chain	$\approx 46 \text{ dB} / \approx 60 \text{ dB}$
Input: I/V peak per leg, nominal / worst	0.96 V / 1.08 V
Input: differential peak, nominal / worst	1.93 V / 2.17 V

5.1 Simulation

AC sweep of one channel with the TI OPA1612 model on $\pm 3.3 \text{ V}$; differential drive of 0.5 V AC on each input, 180° apart; 10 Hz to 10 MHz, 100 points/decade. The I/V stage is not included.

Quantity	Simulated	Hand calculation
Passband gain	-7.1 dB	-7.1 dB (0.441)
-3 dB frequency	69.30 kHz	69.4 kHz
Passband shape	Flat through 20 kHz, no peaking	$Q = 0.707$
Stopband slope	40 dB/decade	40 dB/decade
Level at 1 MHz	-53 dB (46 dB below passband)	46 dB below passband
High-frequency floor	Flattens near -84 dB at 5 to 10 MHz (op-amp gain-bandwidth limit)	Not modelled

6. Open Problems

None.

7. Revision Log

Rev	Date	Change
0	2026-09-24	Initial issue: topology, requirements, component values, equations.
1	2026-09-24	Standard values: $R2 = 681\ \Omega$, $C1 = 13\ \text{nF}$, $C2 = 2.7\ \text{nF}$.
2	2026-09-24	AC-sweep simulation added.
3	2026-09-24	Part numbers added; all resistors 0.1% thin film.
4	2026-09-24	Leg assignment and output polarity recorded.
5	2026-09-25	Reformatted; op-amp in title; requirements and constraints merged; equations typeset.

8. Sources

1. [Analog Engineer's Circuit Cookbook: Amplifiers, MFB Low-Pass Filter \(Texas Instruments\)](#)
2. [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#)
3. [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#)